Supplementary Information
Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024

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Supplementary Note 1. Legal audit and policy-feature coding

The legal audit records one row per state and distinguishes clean source-verified within-panel adopters from verified non-adopters, partial cases, ambiguous cases, and baseline-permitless states. Partial, ambiguous, and baseline-permitless rows are documented so that they are not silently absorbed into the binary treatment definition.

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

Among the 26 clean source-verified adopter rows, 21 had a training requirement removed, 25 removed the carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These policy-feature fields describe the carry-permit process and are treated as exploratory.

Mechanism field/value	State count
Training requirement removed: yes	21
Training requirement removed: no prior training requirement	5
Carry-permit background-check screen removed	25
Carry-permit screen removed while purchase permit retained	1
Permit-specific misdemeanor screen removed	12
Eligibility standard retained but no pre-carry check	7
Other misdemeanor-screen coding categories	7

The full state-level audit below reports the adoption year, audit status, whether the state enters the primary processed treatment map, the coding note, and source URLs used for legal verification. This table is included to make Arkansas, Mississippi, and Vermont visible rather than hiding the edge cases inside a binary treatment variable.

Table 1: State-by-state permitless concealed-carry treatment coding audit.

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Alabama	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/AL/text/HB272/id/2544995; https://www.handgunlaw.us/documents/Permitless˙Ca
Alaska	2003	clean adopter	Yes	Clean source-verified within-panel adopter.	https://dps.alaska.gov/Statewide/R-I/PermitsLicensing/Firearm-FAQ; https://www.handgunlaw.us/documents/Permitless˙Ca
Arizona	2010	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.azleg.gov/legtext/49leg/2r/bills/sb1108s. https://www.handgunlaw.us/documents/Permitless˙Ca

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Arkansas	–	ambiguous reviewed	No	Excluded from primary clean-adoption map; 2021 and 2023 codings tested as sensitivities.	https://arkleg.state.ar.us/Bills/Detail?ddBienniumSess https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
California	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://oag.ca.gov/firearms/ccw
Colorado	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://cbi.colorado.gov/firearms/concealed-handgun-permit-chp-reciprocity
Connecticut	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://portal.ct.gov/despp/division-of-state-police/special-licensing-and-firearms/pistol-permits
Delaware	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://attorneygeneral.delaware.gov/criminal/conceal carry-deadly-weapons-ccdw/
Florida	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.flgov.com/eog/news/press/2023/governor ron-desantis-signs-hb-543-constitutional-carry; https://www.handgunlaw.us/documents/Permitless˙Ca
Georgia	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://gov.georgia.gov/press-releases/2022-04-13/gov-kemp-signs-georgia-constitutional-carry-act-law; https://www.handgunlaw.us/documents/Permitless˙Ca
Hawaii	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://www.honolulupd.org/police-services/firearms/
Idaho	2016	clean adopter	Yes	Clean source-verified within-panel adopter.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.handgunlaw.us/documents/Permitless˙Ca
Illinois	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://isp.illinois.gov/Foid/Ccl
Indiana	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://events.in.gov/event/gov˙holcomb˙issues˙stateme https://www.in.gov/isp/files/Permitless-Carry-Website-Messaging.pdf
Iowa	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.legis.iowa.gov/legislation/billTracking/bil https://www.handgunlaw.us/documents/Permitless˙Ca
Kansas	2015	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.kslegislature.gov/li˙2016/b2015˙16/measu https://www.handgunlaw.us/documents/Permitless˙Ca
Kentucky	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.kentuckystatepolice.ky.gov/ccdw-home/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/
Louisiana	2024	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legis.la.gov/Legis/BillInfo.aspx?i=245605; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Maine	2015	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.maine.gov/dps/msp/licenses-permits/concealed-carry-maine; https://legislature.maine.gov/legis/bills/display˙ps.asp
Maryland	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://mdsp.maryland.gov/Organization/Pages/Crim
Massachusetts	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://www.mass.gov/how-to/apply-for-a-firearms-license
Michigan	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://www.michigan.gov/msp/services/ccw
Minnesota	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://dps.mn.gov/divisions/bca/bca-divisions/administrative/Pages/Permit-to-Carry.aspx
Mississippi	2015	partial	Yes	Retained in processed panel as 2015 adoption; legal audit marks the row partial.	https://billstatus.ls.state.ms.us/documents/2016/html/ 0799/HB0786PS.htm; https://www.handgunlaw.us/documents/Permitless˙Ca
Missouri	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.senate.mo.gov/BillTracking/Bills/BillInfo https://www.handgunlaw.us/documents/Permitless˙Ca
Montana	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/MT/bill/HB102/2021; https://www.handgunlaw.us/documents/Permitless˙Ca
Nebraska	2023	clean adopter	Yes	Clean source-verified within-panel adopter.	https://nebraskalegislature.gov/bills/view˙bill.php?Doc https://statepatrol.nebraska.gov/services/concealed-handgun-permits/chp-frequently-asked-questions
Nevada	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://rccd.nv.gov/FeesForms/Brady/CCW-Permit-Recognition/
New Hampshire	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/NH/votes/SB12/2017; https://www.handgunlaw.us/documents/Permitless˙Ca
New Jersey	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://www.nj.gov/njsp/firearms/forms.shtml
New Mexico	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://www.dps.nm.gov/concealed-carry/
New York	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://gunsafety.ny.gov/pistol-permit-recertification

State	Year	Audit status	Primary map	Coding note	Source URL(s)
North Carolina	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://ncdoj.gov/law-enforcement-training/criminal-justice/officer-certification-programs/concealed-weapon-reciprocity/
North Dakota	2017	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.nd.gov/ndhp/node/116; https://legiscan.com/ND/bill/HB1169/2017
Ohio	2022	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.legislature.ohio.gov//download?key=1896 https://www.handgunlaw.us/documents/Permitless˙Ca
Oklahoma	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.oklegislature.gov/cf˙pdf/2019-20“%20SUPPORT“%20DOCUMENTS/BILLSUM/Hou https://www.handgunlaw.us/documents/Permitless˙Ca
Oregon	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://www.oregon.gov/osp/programs/cjis/pages/oreg concealed-handgun-license.aspx
Pennsylvania	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/state-laws/concealed-carry-in-pennsylvania/; https://giffords.org/lawcenter/state-laws/open-carry-in-pennsylvania/; https://www.pa.gov/en/agencies/psp/programs/firearm firearms-in-pennsylvania.html
Rhode Island	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/open-carry/; https://riag.ri.gov/i-want/get-concealed-carry-permit
South Carolina	2024	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sled.sc.gov/constitutional-carry-guidance; https://legiscan.com/SC/supplement/H3594/id/47310
South Dakota	2019	clean adopter	Yes	Clean source-verified within-panel adopter.	https://sdlegislature.gov/Session/Bill/9374; https://www.handgunlaw.us/documents/Permitless˙Ca
Tennessee	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.wvlt.tv/2021/04/08/governor-lee-signs-permitless-carry-into-law/; https://www.handgunlaw.us/documents/Permitless˙Ca
Texas	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://www.lrl.texas.gov/legis/billsearch/billdetails.cf https://sll.texas.gov/faqs/permitless-carry-guns/
Utah	2021	clean adopter	Yes	Clean source-verified within-panel adopter.	https://bci.utah.gov/concealed-firearm/; https://www.handgunlaw.us/documents/Permitless˙Ca
Vermont	pre-panel	baseline permitless	No	Baseline permitless before the panel; not a within-panel adoption event.	https://www.nraila.org/gun-laws/state-gun-laws/vermont/; https://www.handgunlaw.us/states/vermont.pdf
Virginia	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://vsp.virginia.gov/services/firearms/resident-concealed-handgun-permits/
Washington	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://dol.wa.gov/professional-licenses/concealed-pistol-license
West Virginia	2016	clean adopter	Yes	Clean source-verified within-panel adopter.	https://wvpress.org/breaking-news/wvs-no-permit-concealed-carry-begins-today/; https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/

State	Year	Audit status	Primary map	Coding note	Source URL(s)
Wisconsin	–	verified non-adopter	No	No statewide permitless concealed-carry adoption through 2024.	https://giffords.org/lawcenter/gun-laws/policy-areas/guns-in-public/concealed-carry/; https://www.usconcealedcarry.com/blog/what-is-open-carry-and-which-states-allow-it/; https://www.wisdoj.gov/Pages/PublicSafety/concealed carry-weapon-license.aspx
Wyoming	2011	clean adopter	Yes	Clean source-verified within-panel adopter.	https://legiscan.com/WY/bill/SF0047/2011; https://www.handgunlaw.us/documents/Permitless˙Ca

Supplementary Note 2. Main and secondary mortality estimates

Outcome	Coef.	SE	p	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Descriptive outcome trends are included as a supporting check, not as causal evidence. They show why the paper separates suicide-related outcomes from homicide outcomes before fitting adjusted models.

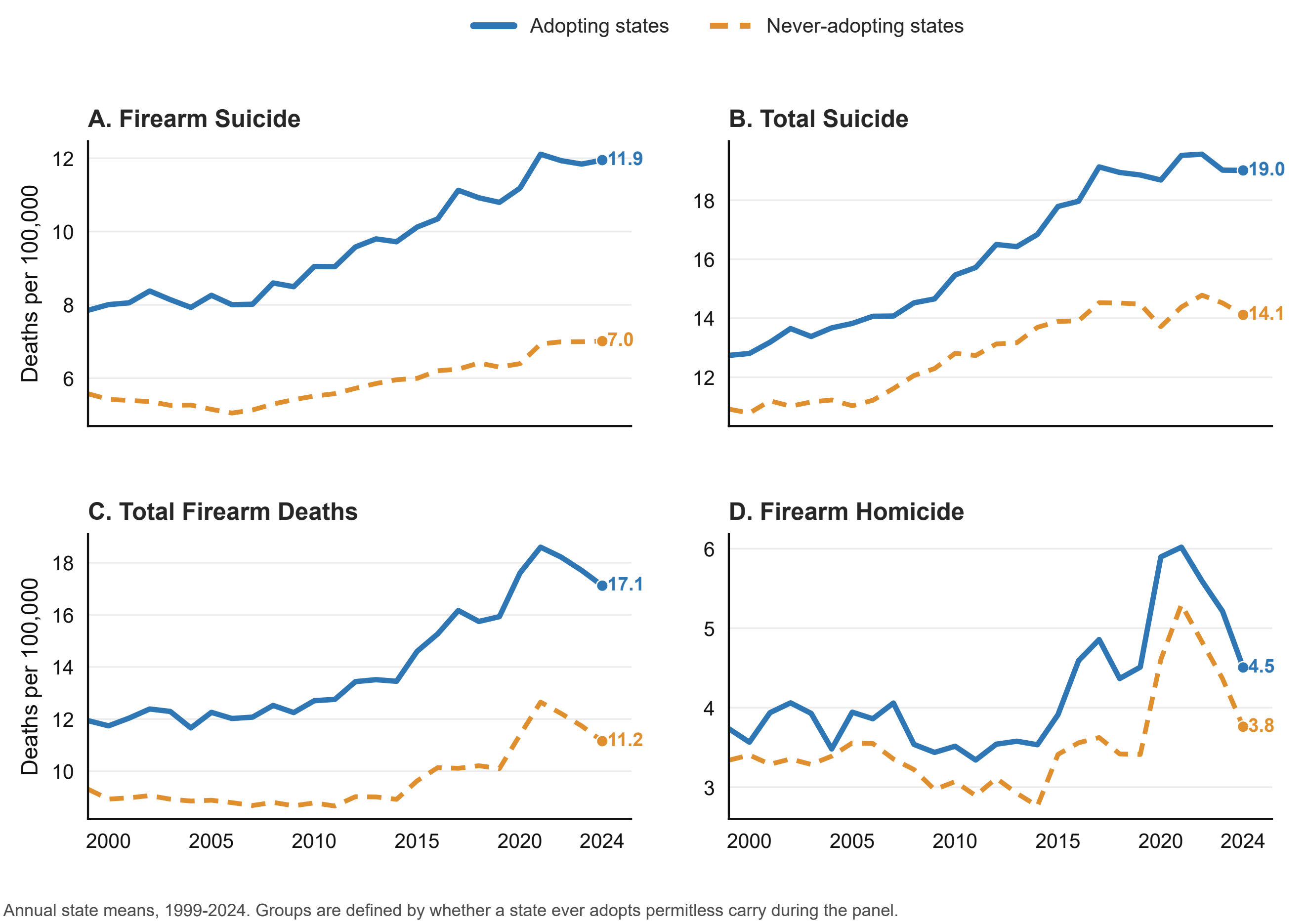

Figure 1: **Descriptive outcome trends by permitless-carry adoption status.** Firearm-suicide and total-suicide rates were higher in adopter states, while homicide trends were noisier and did not visually follow the same pattern. The figure is descriptive because adopter and non-adopter states differed before adoption.

Supplementary Note 3. Firearm-suicide specification and timing summary

The summary places the baseline fixed-effects estimate beside population weighting, COVID-window restrictions, leave-one-adopter-out checks, placebo timing, state-specific trends, cohort-window ATT estimates, and event-time diagnostics. It is the main table for interpreting direction, magnitude sensitivity, and identification fragility.

Table 2: **Firearm-suicide specification and timing summary.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	$1.391 > 0.338$	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	Alternative staggered-adoption estimand
Not-yet-treated dynamic ATT	0.387	–	Alternative timing comparison
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Cohort-window and not-yet-treated ATT rows are alternative timing estimands. The event-time row is a pretrend diagnostic; its p column reports the minimum pre-adoption p rather than a post-adoption effect test.

The focused robustness table below reports the estimates used in the main-text robustness paragraph: binary TWFE, fractional-year TWFE, stacked DiD, covariate-balanced TWFE, and firearm suicide minus non-firearm suicide.

Table 3: **Primary firearm-suicide robustness estimates.**

Specification	Estimate	SE	p	N	Interpretation
Binary TWFE	1.391	0.222	< 0.001	1222	Headline annual treatment coding
Fractional-year TWFE	1.534	0.239	< 0.001	1222	Effective-year exposure prorated by law date
Stacked DiD	0.665	0.137	< 0.001	3446	Cohort stacks with not-yet-treated controls
Covariate-balanced TWFE	0.889	0.304	0.003	1222	Non-adopters reweighted toward adopter covariates
Firearm minus non-firearm suicide TWFE	0.976	0.226	< 0.001	1222	Firearm-specific association net of non-firearm suicide movement

Estimates are deaths per 100,000 residents. The first four rows estimate firearm suicide. The final TWFE row estimates firearm suicide minus non-firearm suicide, so it is a firearm-specific check rather than a total-suicide effect. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. The stacked DiD row uses 5-year pre/post cohort stacks with not-yet-treated controls.

A null-imposed Rademacher wild-cluster bootstrap with 999 replications over 47 state clusters yielded $p = 0.001$ for the baseline firearm-suicide row. The same bootstrap gave $p = 0.016$ for total firearm deaths and $p = 0.509$ for firearm homicide. These values address small-cluster inference for the baseline TWFE rows; they do not address the separate pretrend, trend-specification, or common-support limitations.

Total firearm deaths are retained as a secondary outcome-family check rather than a headline finding. The baseline total-firearm estimate was positive (1.271; $p = 0.008$), but it attenuated to 0.070 under state-specific trends ($p = 0.806$) and 0.141 under covariate balancing ($p = 0.851$).

Supplementary Note 4. Age-adjusted outcome-rate sensitivity

Crude mortality rates are the primary outcome scale because they measure realized state-year mortality burden. Age-adjusted rates are reported as a demographic-composition sensitivity check using CDC WONDER age-adjusted rates standardized to the 2000 U.S. standard population. The age-adjusted estimates preserve the crude-rate pattern: firearm suicide, total suicide, and total firearm deaths remain positive, while firearm homicide remains statistically weak and centered below zero. The total-firearm row is not interpreted as a stable headline result because later trend and covariate-balancing checks attenuate it.

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

Supplementary Note 5. Firearm-law and contextual covariate sensitivity

External firearm-law covariates, health-access covariates, and demographic-poverty covariates attenuated but did not erase the firearm-suicide estimate. Overdose controls reduced precision in a short sample. The short HRSA mental-health-provider window did not retain statistical significance and should be interpreted as a short-window sensitivity rather than a full-panel replacement. These rows are covariate-layer sensitivity checks, not proof that adding more time-varying covariates necessarily improves identification.

Firearm-suicide specification	Coef.	p	N
Baseline fixed effects	1.391	< 0.001	1222
External firearm-law covariates	1.096	< 0.001	1222
Health-access covariates	0.707	< 0.001	752
Overdose covariates	0.492	0.060	282
Demographic-poverty covariates	0.695	< 0.001	940
Mental-health-provider access covariates	0.116	0.744	141
Full Phase 3B2 covariates	0.176	0.633	141

Supplementary Note 6. Change-score checks

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.658	0.634	0.699
Non-firearm suicide	0.214	0.134	0.047
Total suicide	0.872	0.767	0.746
Firearm homicide	-0.034	0.024	0.035
Total firearm deaths	0.537	0.617	0.718

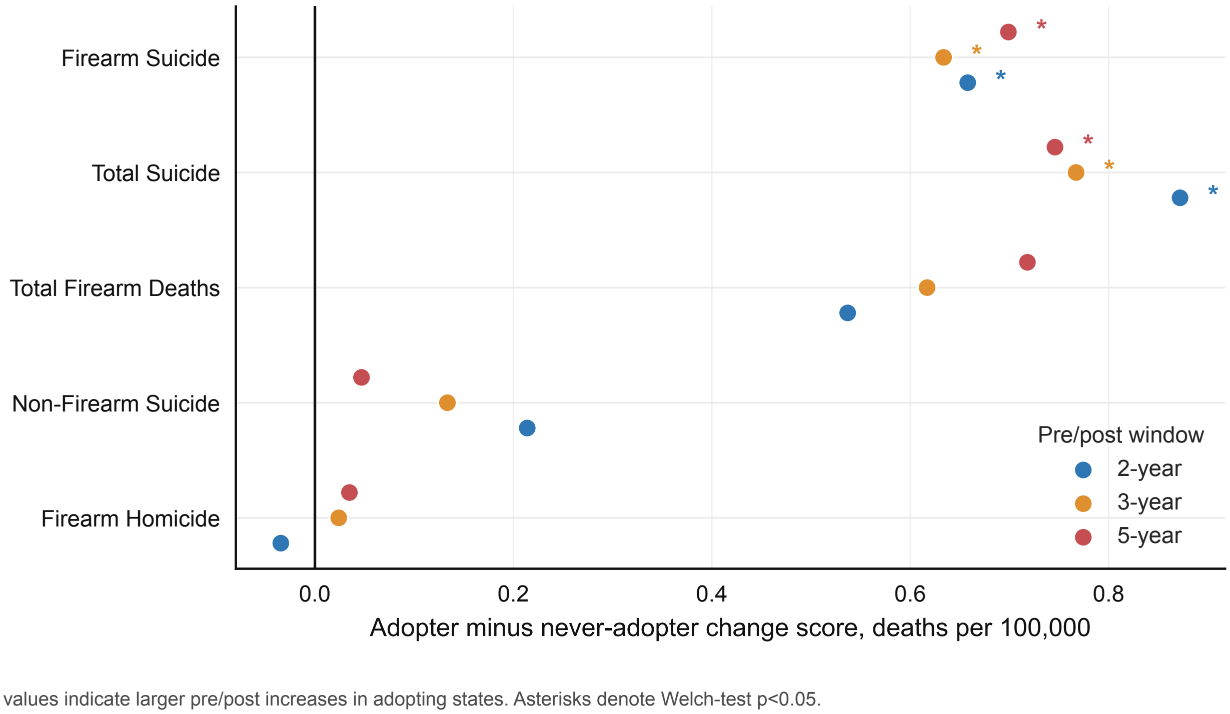

Figure 2: **Change-score robustness across pre-post windows.** Firearm suicide is positive across 2-, 3-, and 5-year windows. Firearm homicide remains statistically weak across the same windows. These window checks support the suicide-homicide contrast but are less adjusted than the fixed-effects models.

Supplementary Note 7. Staggered-adoption estimates and event-time diagnostics

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive, smaller than TWFE.
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive, smaller than TWFE.
Total suicide	0.871	0.816	0.882	0.464	Positive, with pretrend flag.
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive estimate.
Total firearm deaths	0.382	0.435	0.493	0.200	Positive, with pretrend flag.

The stacked DiD specification estimated a firearm-suicide association of 0.665 deaths per 100,000 (SE 0.137; $p < 0.001$). This estimate is smaller than the baseline binary TWFE estimate and closer to the cohort-window ATT rows, which supports the manuscript's interpretation that the direction is more stable than the exact magnitude.

Supplementary Note 8. Robustness and Arkansas sensitivity

Outcome	Baseline	P05 specs	Leave-one min	Leave-one max	Flag
Firearm suicide	1.391	5	1.265	1.481	Stable positive, trend-sensitive.
Non-firearm suicide	0.415	5	0.371	0.472	Stable positive, trend-sensitive.
Total suicide	1.805	6	1.656	1.954	Stable positive.
Firearm homicide	-0.234	0	-0.399	-0.049	No stable positive signal.
Total firearm deaths	1.271	4	1.058	1.532	Positive, trend-sensitive.

Arkansas recoding as 2021 or 2023 retained the sign of all five outcomes. Firearm suicide, total suicide, non-firearm suicide, and total firearm deaths retained $p < 0.05$ across both Arkansas codings; firearm homicide did not. Reintroducing Mississippi alone and in combination with both Arkansas codings preserved the positive firearm-suicide and total-suicide pattern.

Table 4: **Firearm-suicide leave-one-adopter-out estimates by excluded state.**

Excluded state	Coef.	p
Alabama	1.415	< 0.001
Alaska	1.405	< 0.001
Arizona	1.481	< 0.001
Florida	1.398	< 0.001
Georgia	1.413	< 0.001
Idaho	1.375	< 0.001
Indiana	1.393	< 0.001
Iowa	1.395	< 0.001
Kansas	1.414	< 0.001
Kentucky	1.451	< 0.001
Louisiana	1.411	< 0.001
Maine	1.399	< 0.001
Missouri	1.366	< 0.001
Montana	1.342	< 0.001
Nebraska	1.404	< 0.001
New Hampshire	1.391	< 0.001
North Dakota	1.358	< 0.001
Ohio	1.410	< 0.001
Oklahoma	1.383	< 0.001
South Carolina	1.406	< 0.001
South Dakota	1.403	< 0.001
Tennessee	1.416	< 0.001
Texas	1.415	< 0.001
Utah	1.424	< 0.001
West Virginia	1.420	< 0.001
Wyoming	1.265	< 0.001

Table 5: **Firearm-suicide never-treated placebo timing estimates.**

Placebo shift	Coef.	p
0	-0.026	0.850
1	-0.246	0.173
2	-0.062	0.788
3	0.081	0.721
4	0.385	0.060
5	0.291	0.047
6	0.054	0.675
7	0.050	0.767
8	-0.032	0.883
9	-0.224	0.375
10	-0.275	0.156

The absolute median placebo estimate was 0.081 deaths per 100,000 and the absolute 95th percentile was 0.338. The observed firearm-suicide estimate was 1.391, exceeding that placebo timing distribution.

Supplementary Note 9. Heterogeneity and policy-feature checks

Rurality was the clearest recurring heterogeneity dimension: firearm suicide, non-firearm suicide, and total suicide had positive high-rurality interactions. These models are exploratory because state-level subgroup indicators are coarse and adoption is politically patterned.

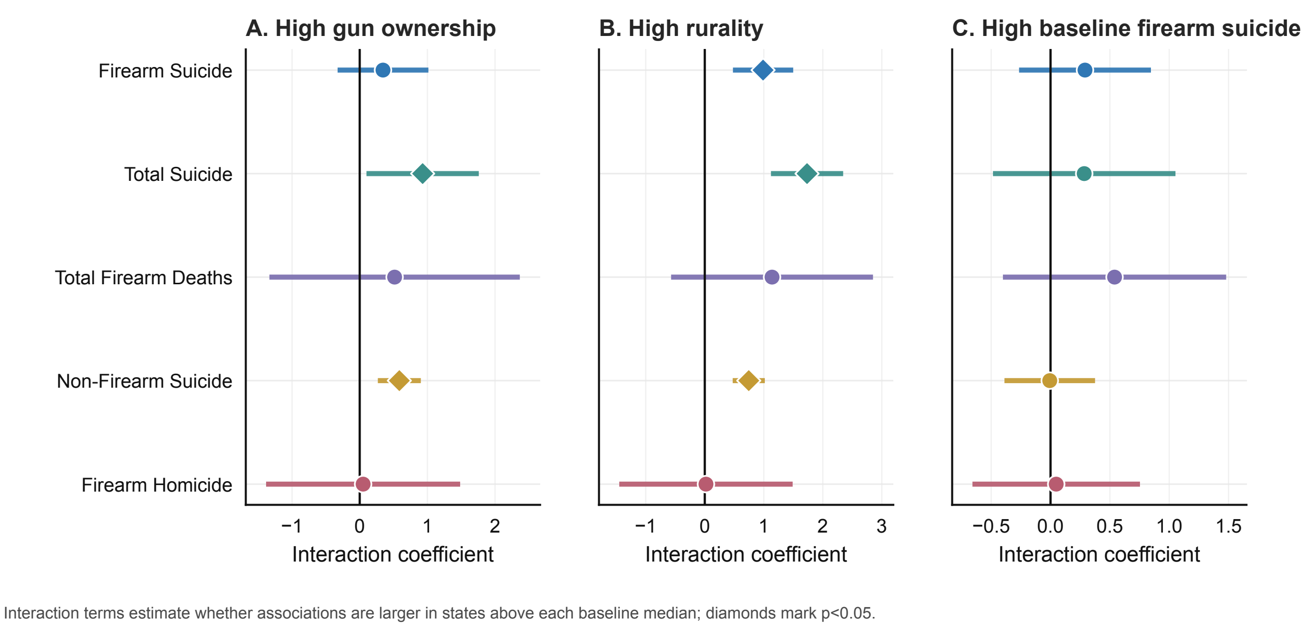

Figure 3: **Exploratory heterogeneity checks.** Rurality is the clearest recurring subgroup pattern for suicide-related outcomes. These results describe where the association was larger; they do not identify the mechanism.

Outcome/dimension	Interaction	p
Firearm suicide, high rurality	0.988	< 0.001
Total suicide, high rurality	1.733	< 0.001
Non-firearm suicide, high rurality	0.745	< 0.001
Non-firearm suicide, high gun ownership	0.586	< 0.001

Policy-feature heterogeneity did not identify a stable mechanism. None of the 15 training, carry-permit background-check, or misdemeanor-screen interaction rows had $p < 0.05$. Five rows were sparse or unavailable because the feature-specific clean-adopter comparison group was absent.

Supplementary Note 10. NICS mechanism proxy and additional CDC WONDER checks

The NICS mechanism proxy uses the official FBI Year by State/Type table where local text extraction produced reliable state-year rows. The processed panel covers 50 states from 1999 through 2021. The direct handgun-check proxy increased after permitless-carry adoption, but the permit and combined handgun-or-permit rows do not show a net increase. These rows are supplemental because NICS checks are not firearm sales and because the locally parsed official file did not produce clean 2022–2024 state-year rows.

NICS proxy	Sample	Coef.	SE	p
Handgun checks per 100,000	1999–2021	476.952	185.444	0.010
Permit checks per 100,000	1999–2021	-2274.542	1846.033	0.218
Permit plus recheck checks per 100,000	1999–2021	-643.925	1240.223	0.604
Handgun or permit checks per 100,000	1999–2021	-1797.590	1874.740	0.338
Handgun checks per 100,000	Excluding 2020–2021	379.432	175.813	0.031
Handgun or permit checks per 100,000	Excluding 2020–2021	-976.463	1183.758	0.409

The negative-control mortality checks do not show a uniform positive mortality shift. Cancer mortality was positive but statistically weak, cardiovascular mortality was near zero, and motor-vehicle mortality moved in the opposite direction. The added injury and poisoning checks were mixed: fall mortality was positive but statistically weak, other non-transport injury excluding falls and accidental poisoning was small and negative, and accidental-poisoning mortality was negative and imprecise. The other non-transport row excludes both falls and accidental poisoning so the injury rows are not mechanically overlapping; the accidental-poisoning row directly tests the overdose/despair pathway. This does not make adopter and non-adopter states exchangeable, but it weakens the interpretation that the firearm-suicide estimate is merely a uniform rise across all mortality outcomes.

Table 6: **Negative-control mortality estimates.**

Outcome	Coef.	SE	p	N
Cancer mortality	1.796	2.332	0.441	1222
Cardiovascular mortality	-0.591	5.810	0.919	1222
Motor-vehicle mortality	-1.319	0.416	0.002	1222

Table 7: **Additional injury and accidental-poisoning mortality estimates.**

Outcome	Coef.	SE	p	N
Fall mortality	1.178	0.913	0.197	1222
Other non-transport injury excluding falls/poisoning	-0.218	0.285	0.445	1222
Accidental poisoning mortality	-1.484	1.943	0.445	1215

Sex/age stratification shows that the firearm-suicide association is concentrated among males. Female firearm-suicide estimates were small and statistically weak, while male estimates were positive overall and across broad adult age groups. These rows are state-level stratified outcomes, not individual-level mechanism evidence. The analysis uses complete broad-age cells after CDC WONDER suppression, so observation counts differ across strata.

Table 8: **Firearm-suicide estimates by sex and broad age group.**

Stratum	Coef.	SE	p	N
Female, modeled adult ages	0.108	0.200	0.590	749
Male, modeled adult ages	1.665	0.389	< 0.001	1141
Male, age 15-24	1.651	0.760	0.030	1085
Male, age 25-44	1.702	0.483	< 0.001	1038
Male, age 45-64	1.830	0.495	< 0.001	1066
Male, age 65+	1.332	0.737	0.071	631

The suppression audit below reports the number of clean-sample state-years retained after requiring complete broad-age cells. Adult male strata retain most clean-sample state-years, while female strata and the male 65+ row are more affected by WONDER suppression. The sex/age results should therefore be read as an epidemiologic coherence check rather than as equally powered subgroup estimates.

Table 9: **CDC WONDER sex/age suppression audit.**

Stratum	Modeled clean cells	Expected clean cells	Share	Complete WONDER cells
Female, age 15-24	380	1222	31.1%	424
Female, age 25-44	386	1222	31.6%	419
Female, age 45-64	461	1222	37.7%	499
Female, age 65+	151	1222	12.4%	187
Male, age 15-24	1085	1222	88.8%	1140
Male, age 25-44	1038	1222	84.9%	1092
Male, age 45-64	1066	1222	87.2%	1124
Male, age 65+	631	1222	51.6%	652

Supplementary Note 11. Homicide missingness audit

Firearm-homicide outcome rows have fewer observations than suicide and total-firearm rows because some state-year rates are unavailable or suppressed. The missingness audit shows that all missing firearm-homicide rows occur in the smallest-population state-years.

Table 10: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

State	Missing	Possible	Missing pct.
New Hampshire	16	26	61.5%
North Dakota	15	26	57.7%
South Dakota	14	26	53.8%
Wyoming	14	26	53.8%
Hawaii	13	26	50.0%
Maine	6	26	23.1%
Delaware	1	26	3.8%
Montana	1	26	3.8%

Supplementary Note 12. Full diagnostic source files

The following repository outputs provide the full machine-readable diagnostic trail used to write the manuscript and this supplement:

- `outputs/tables/did/twfe_did_main_results.csv`
- `outputs/tables/did/twfe_did_ageadj_sensitivity_summary.csv`
- `outputs/tables/did/twfe_did_firearm_law_control_summary.csv`
- `outputs/tables/did/twfe_did_nonfirearm_confounder_summary.csv`
- `outputs/tables/did/twfe_did_phase3b2_confounder_summary.csv`
- `outputs/tables/main/welch_change_score_results.csv`
- `outputs/tables/modern_did/modern_did_summary.csv`
- `outputs/tables/modern_did/stacked_did_results.csv`
- `outputs/tables/modern_did/event_time_never_control_estimates.csv`
- `outputs/tables/robustness/robustness_summary.csv`
- `outputs/tables/robustness/fractional_timing_results.csv`
- `outputs/tables/robustness/covariate_balanced_twfe_results.csv`
- `outputs/tables/robustness/covariate_balance_diagnostics.csv`
- `outputs/tables/robustness/firearm_suicide_main_robustness_table.csv`
- `outputs/tables/robustness/leave_one_state_out.csv`
- `outputs/tables/robustness/placebo_never_treated.csv`
- `outputs/tables/robustness/arkansas_treatment_sensitivity_summary.csv`
- `outputs/tables/heterogeneity/heterogeneity_did_results.csv`
- `outputs/tables/heterogeneity/firearm_suicide_sex_age_twfe_results.csv`
- `outputs/tables/heterogeneity/firearm_suicide_sex_age_suppression_audit.csv`
- `outputs/tables/negative_controls/negative_control_twfe_results.csv`
- `outputs/tables/mechanism/mechanism_heterogeneity_results.csv`
- `outputs/tables/mechanism/nics_handgun_proxy_results.csv`
- `outputs/tables/mechanism/policy_bundle_distribution.csv`
- `outputs/tables/data_availability/additional_analysis_data_availability.csv`

- data/processed/analysis_panel_negative_control_mortality_1999_2024.csv
- data/processed/analysis_panel_firearm_suicide_by_sex_age_1999_2024.csv
- data/policy/permitless_carry_legal_audit.csv